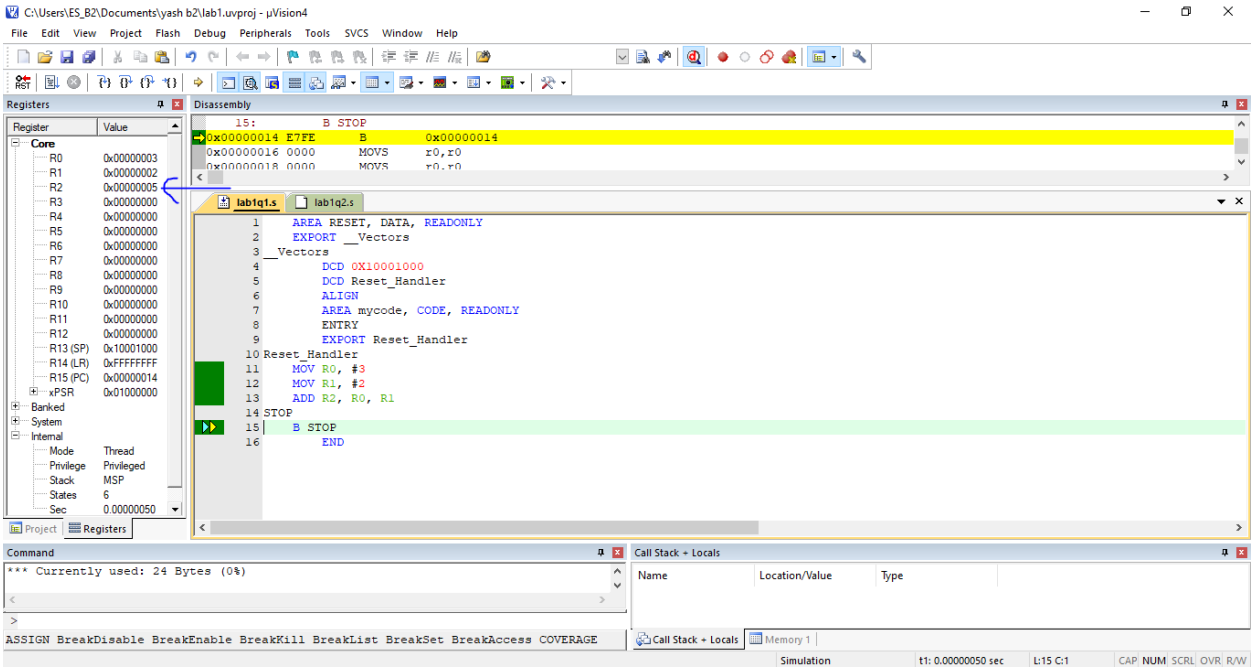
Name: Yashvardhan Gupta

Embedded Systems LAB 1

Q1 Write a program to add contents of two registers and store them into R2.

```
        AREA RESET, DATA, READONLY
        EXPORT __Vectors
__Vectors
                DCD 0X10001000
                DCD Reset_Handler
                ALIGN
                AREA mycode, CODE, READONLY
                ENTRY
                EXPORT Reset_Handler
Reset_Handler
        MOV R0, #3
        MOV R1, #2
        ADD R2, R0, R1
STOP
        B STOP
                END
```

Output



Here we can see that the blue arrow points to register **R2** value which contains the final answer by **adding** the values of registers **R0,R1**. It is in hexadecimal form.

<u>Q2</u> Write a program to add contents of registers R0, R1, R2, R3 and store them into register R10

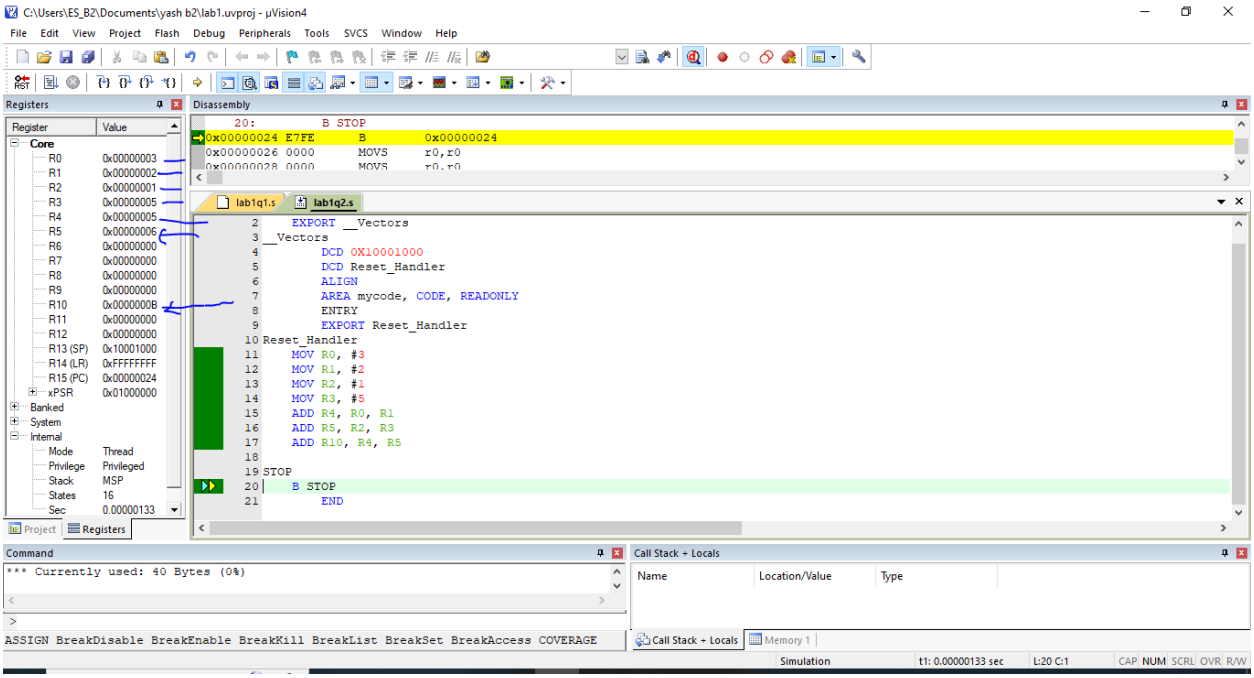
```
AREA RESET, DATA, READONLY
```

```
        EXPORT __Vectors
__Vectors
                DCD 0X10001000
                DCD Reset_Handler
                ALIGN
                AREA mycode, CODE, READONLY
                ENTRY
                EXPORT Reset_Handler
Reset_Handler
        MOV R0, #3
        MOV R1, #2
        MOV R2, #1
        MOV R3, #5
        ADD R4, R0, R1
        ADD R5, R2, R3
        ADD R10, R4, R5

STOP
        B STOP
                END
```

<u>Output</u>

<u>Debug session</u>



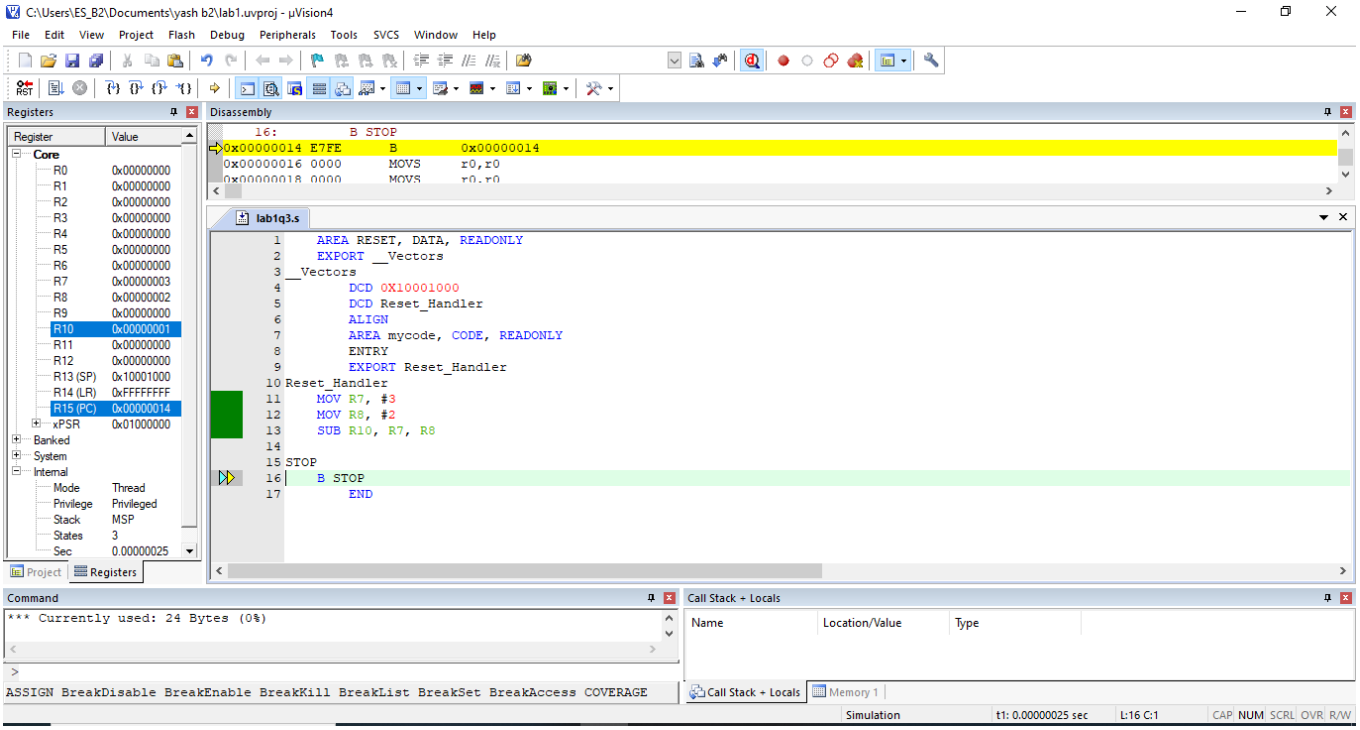
Here we can see that the blue arrow points to register **R10** value which contains the final answer by **adding** the values of registers **R0,R1,R2,R3**. It is in hexadecimal form 0x0000000B which is 11 in decimal.

<u>Q3</u> Write a program to subtract contents of R8 from register R7 and store result in register R10.

```
        AREA RESET, DATA, READONLY
        EXPORT __Vectors
__Vectors
                DCD 0X10001000
                DCD Reset_Handler
                ALIGN
                AREA mycode, CODE, READONLY
                ENTRY
                EXPORT Reset_Handler
Reset_Handler
        MOV R7, #3
        MOV R8, #2
        SUB R10, R7, R8

STOP
        B STOP
                END
```

<u>Output</u>



Contents of **R8(2)** was **subtracted from** register **R7**(3) and stored in register R10(1). These values are in hexadecimal form.

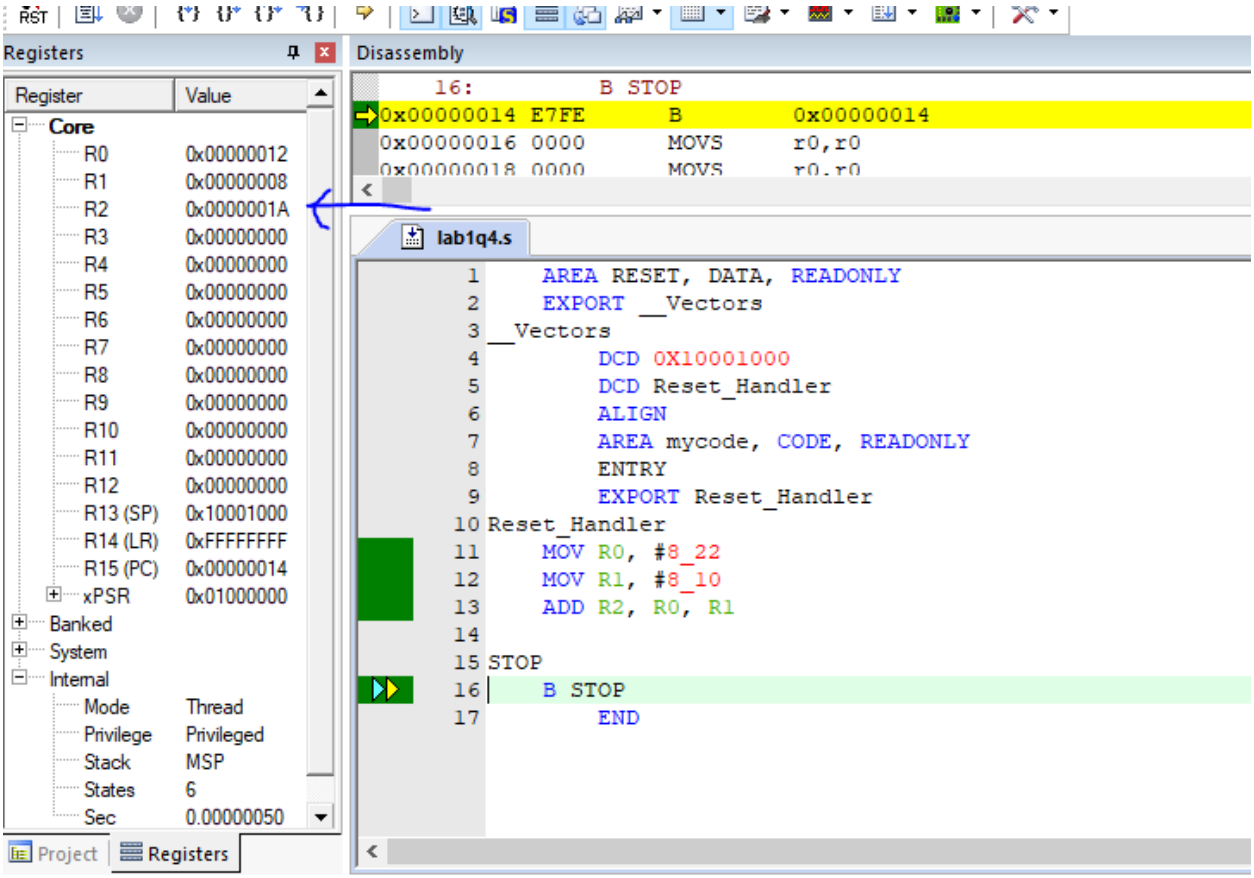
<u>Q4</u> Write a program to add two octal numbers.

```
AREA RESET, DATA, READONLY
EXPORT __Vectors
```

```
__Vectors
        DCD 0X10001000
        DCD Reset_Handler
        ALIGN
        AREA mycode, CODE, READONLY
        ENTRY
        EXPORT Reset_Handler
Reset_Handler
    MOV R0, #8_22
    MOV R1, #8_10
    ADD R2, R0, R1

STOP
    B STOP
        END
```

<u>Output</u>



Register R0 has 22 in octal form which is 18 in decimal form.

Register R1 has 10 in octal form which is 8 in decimal form.

Contents of registers **R0** and **R1** are **added** and stored in register **R2**.

Blue arrow points to register R2 value which shows 1A in hexadecimal.

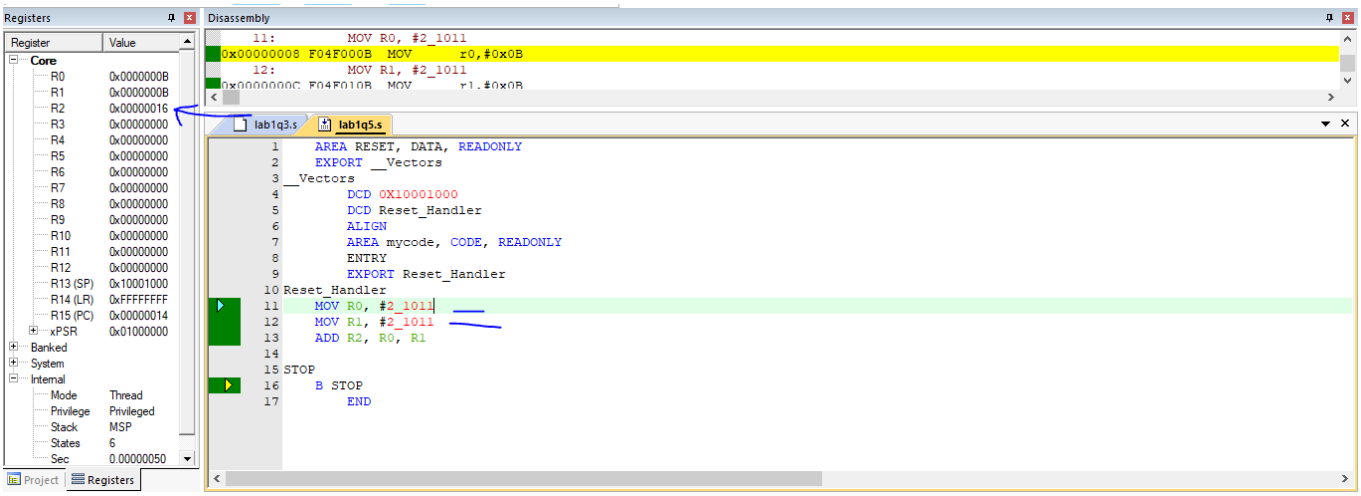
1A in hexadecimal is 26 in decimal.

<u>Q5</u> Write a program to add two binary no.s

```
        AREA RESET, DATA, READONLY
        EXPORT __Vectors
__Vectors
                DCD 0X10001000
                DCD Reset_Handler
                ALIGN
                AREA mycode, CODE, READONLY
                ENTRY
                EXPORT Reset_Handler
Reset_Handler
        MOV R0, #2_1011
        MOV R1, #2_1011
        ADD R2, R0, R1

STOP
        B STOP
                END
```

<u>Output</u>



Here the value of register **R0**(1011 i.e. 11) is added to register **R1**(1011 i.e. 11).

22(10110 or 16 in hexadecimal) is stored in **R2.**